

KWOK KEITH

keithkwok99@gmail.com · +65 84444106

www.linkedin.com/in/kwokkeith · https://kwokkeith.github.io/Portfolio

Education

Singapore University of Technology and Design (SUTD)

Bachelor of Engineering in Computer Science

Singapore

May 2025

- Global Distinguished Scholarship, Lee Kuan Yew Scholarship
- Director's Honor List 2021/22, 2022/23, 2023/24

Reykjavik University

Full-time exchange student

Reykjavik, Iceland

Spring 2024

Stanford University

Full-time exchange student

Palo Alto, CA

Summer 2022

Experience

Hewlett Packard Inc.

Automation Software Developer Intern

Singapore

September 2023 - December 2023

- Developed end-to-end software solutions such as Image to Text application using Tesseract Optical Character Recognition as well as Regulatory Web Documentation using Ruby and Sass Framework.
- Developed a live language translation tool powered by AI for meetings with global stakeholders by bench-marking different models such as WhisperAI, Helsinki, Mistral-OpenOrca and seamlessM4T.

Singapore University of Technology and Design

Undergraduate AI Researcher

Singapore

August 2023 - March 2024

- Worked closely together with a team of 5 researchers to understand the impact of Large Language Models (LLMs) and their effectiveness on student's learning.
- Analyzed and generated insights from qualitative data that has shown various student behaviors and habits when using GPT for their learning.

Project

American Sign Language Translation

Machine Learning

SUTD

January 2025 - April 2025

- Designed a deep neural network incorporating MediaPipe, binary classifier for dynamic or static actions, Long-Short Term Memory, Convolutional Neural Network and ending with a Large Language Model for structured English translations from American Sign Language.

Web Assembly Compiler

Compiler Design

SUTD

September 2024 - December 2024

- A compiler designed to translate SIMP language into WebAssembly, built using Haskell with a Yesod-based frontend.
- Developed the compilation pipeline, including lexing, parsing, static analysis, optimisation, and code generation, delivering efficient and portable WebAssembly output.

Raft Consensus

Software Development

Reykjavik University

April 2024 - May 2024

- Implemented the Raft Consensus algorithm for distributed and concurrent programming in GO.

Finance Automation

Software Development

SUTD

June 2023 - August 2023

- Led, designed and developed an end-to-end full-stack software solution that automates the translation of Receipt-To-Text using Optical Character Recognition Technology (Tesseract)
- Automated transfers of stored receipt details to SUTD SAP Concur Financial System by making use of authorized browser Cookie sessions and HTTP Messages.
- Designed and implemented a database that integrates with the native OS file system implemented with C# with .NET MAUI framework.

Skills

Technical Languages

Python, C#, Java, Javascript, Haskell, GO, Rust, C, C++

Systems and Frameworks

.NET Core, ROS, Git, Linux, React, Unity, FPGA, Lucid HDL